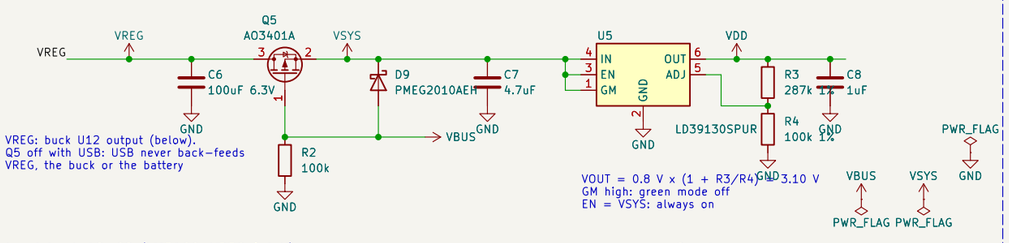
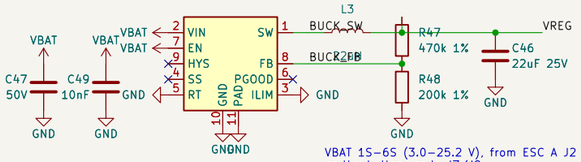
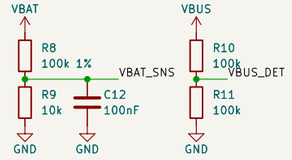
A — Input buck U12 (1S–6S –> VREG 4.1 V), USB/battery power path, LDO



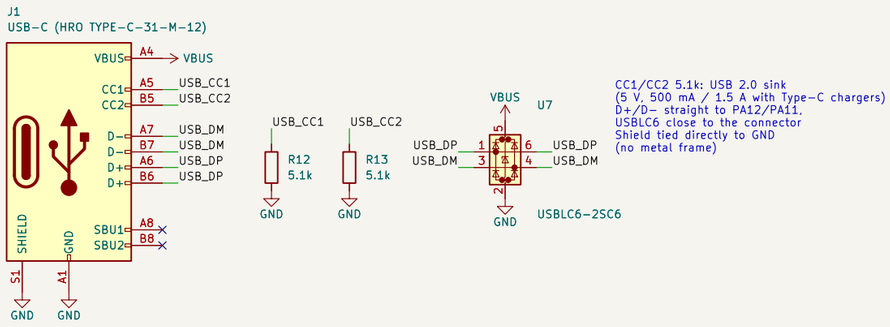
Input buck 1S–6S (LM5166, VREG 4.1 V)



Sensing

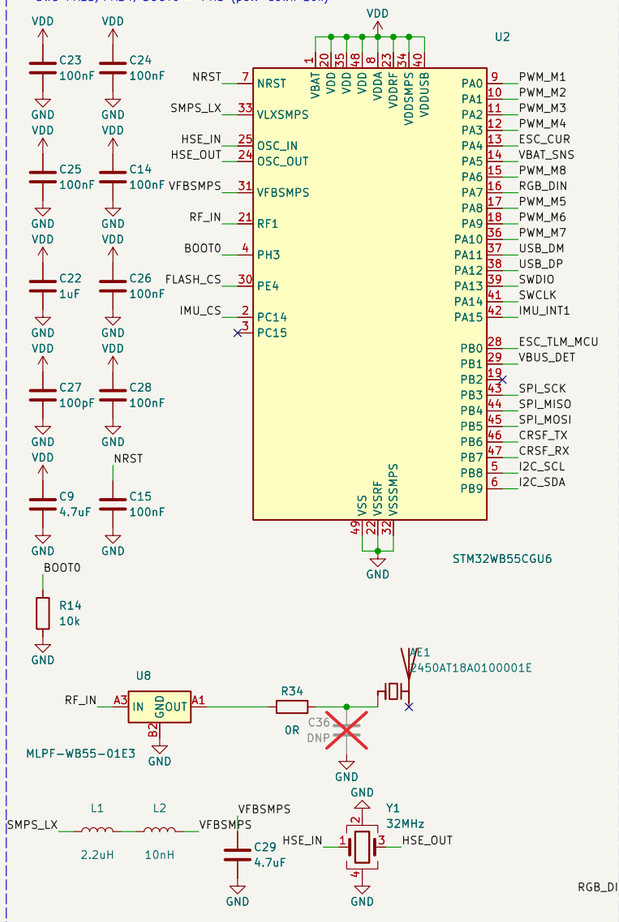


B — USB-C (DFU/CDC, bench supply; no charger), ESD protection

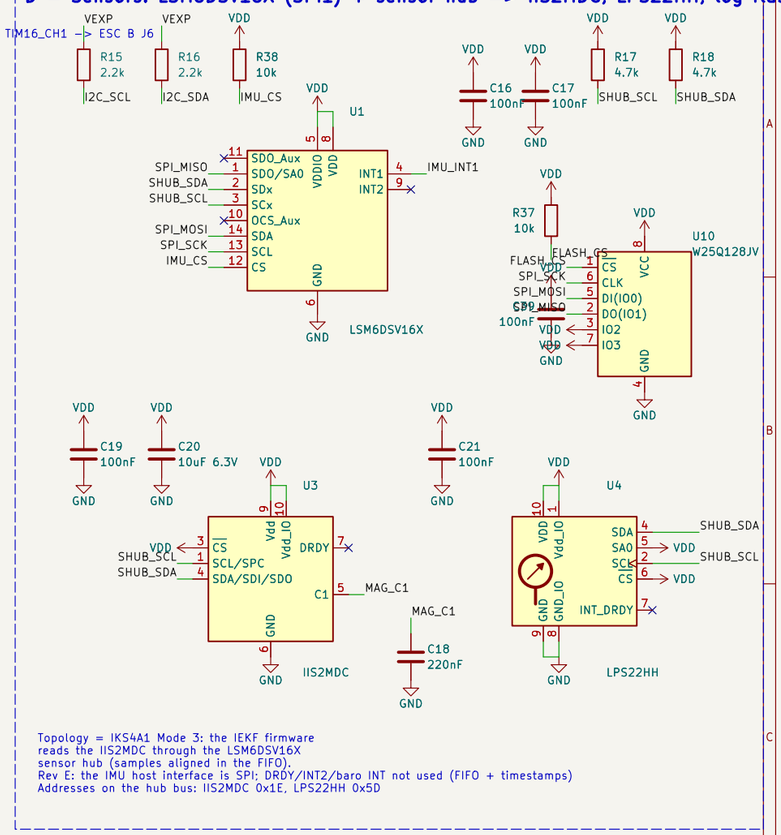


C — STM32WB55CGU6 (QFN48), RF front end, clocks

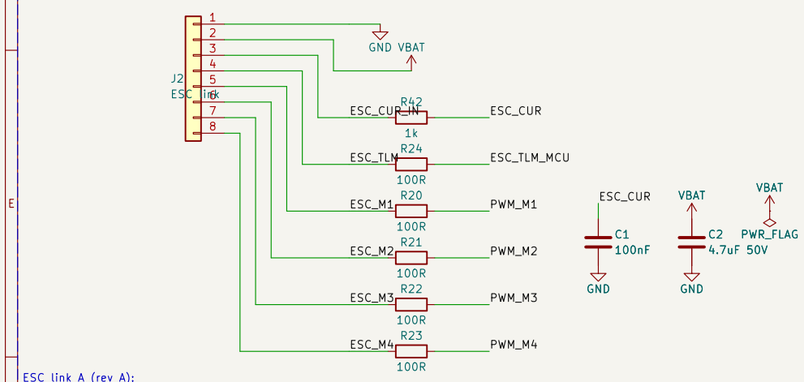
Pin map (IEKF firmware), rev A:
Motors M1–M4 PA0–PA3 TIM2_CH1–CH4 –> ESC A J2; M5–M7 PA8–PA10 TIM1_CH1–CH3, M8 PA6 TIM16_CH1 –> ESC B J6
(PWM 32 kHz brushed / DShot, bidirectional DShot: every motor timer has DMA)
ADC: PA4 ESC_CUR (IN9), PA5 VBAT_SNS (IN10, x10/110); ESC A serial telemetry PB0 (optional)
SPI1 PB3 SCK / PB4 MISO / PB5 MOSI: IMU (CS PC14) + log flash (CS PE4), IMU_INT1 PA15
USART1 PB6 TX / PB7 RX −> ELRS (CRSF 420 kbaud), J5
I2C1 PB8/PB9 −> expansion bus (Qwiic J3 + J4: GPS board, flow board, sensors)
RGB LEDs PA7 (TIM17_CH1 + DMA, buffer U13, 4 x XL-2020); VBUS_DET PB1; PB2 free
No LSE: RF wake-up HSE/1024, RTC HSE/32. USB PA11/PA12 (DFU, CDC)
SWD PA13/PA14, BOOT0 = PH3 (pull-down 10k)



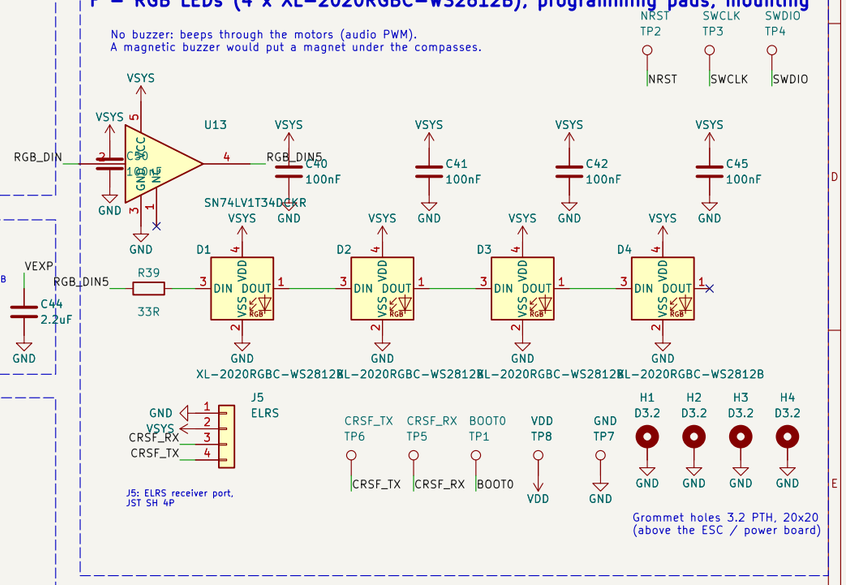
D — Sensors: LSM6DSV16X (SPI1) + sensor hub –> IIS2MDC, LPS22HH; log flash



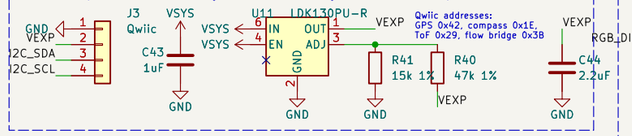
E — ESC link: JST SH 8-pin, common 4-in-1 ESC pinout



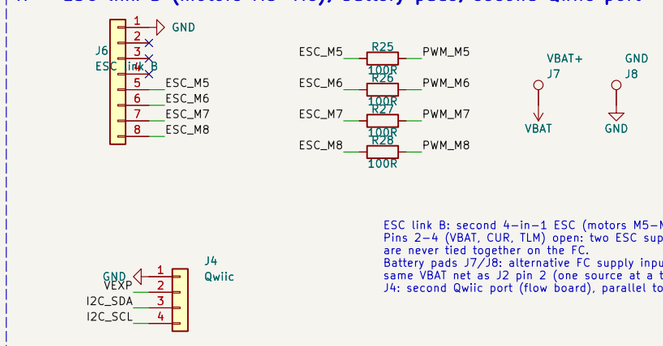
F — RGB LEDs (4 x XL-2020RGBC-WS2812B), programming pads, mounting



G — Expansion bus (Qwiic J3, J4) and ELRS port



H — ESC link B (motors M5–M8), battery pads, second Qwiic port



Generated automatically: run ERC before release
Pin map: ACwing_FC_A_pinmap.md
Power and motors from a 4-in-1 ESC (JST SH 8-pin ESC A J2, ESC B J6) or the ACwing 1S power b
Outline: frame C (40 x 40, as draft E); holes 20 x 20 (mechanical handoff, update 2026-10-18)

ACwing — Andrea Cirillo

Sheet: /		
File: ACwing_FC_A.kicad_sch		
Title: ACwing FC rev A — 1S–6S flight controller (STM32WB55CGU6), 4-in-1		
Size: A3	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/1